

IN-CLASS EXERCISE & HOMEWORK 3

The Campus Delivery Robot

The four tasks of system thinking, applied once, carefully

Group members: _____

Date: _____

Reading: Chapters 1–3 of E. Crawley, B. Cameron, and D. Selva (2015). *System Architecture: Strategy and Product Development for Complex Systems*. Upper Saddle River, NJ: Prentice Hall.

Scenario

A large public university in the upper Midwest has put a small fleet of six-wheeled delivery robots on its campus sidewalks. Each robot is about the size of a large cooler and carries food from a dining hall to a pickup spot outside a residence hall.

One delivery works like this. A student places an order in a phone app. A dining hall worker puts the meal in the robot's insulated cargo bin and closes the lid. The robot drives itself along the sidewalk, crosses one intersection, and stops at the pickup spot. The lid unlocks when the student's phone is close enough. The student takes the meal, and the robot drives back.

The vendor runs an operations center where one person watches the whole fleet on a screen. When a robot gets stuck, and it does get stuck, an operator takes over with a joystick and drives it a few feet until it can continue on its own.

The robots have been running for two semesters. They mostly work. Three things surprised people:

- Around noon, robots and students share the narrow walkway by the library. The robots stop for people, people step aside for the robots, and everyone slows down.
- Students have started walking right behind a robot to get through a crowd.
- In February the grounds crew plowed snow onto the curb ramps. The robots could not get off the sidewalk, and half of campus became unreachable.

What You Are Doing

You are going to run the four tasks of system thinking on this robot, in order, and then answer one question about complexity. This is Chapter 2 and Chapter 3, applied to something you can picture.

Each part starts with a worked example from the textbook so you can see the pattern. Copy the pattern. Short answers are fine, and a sketch is often better than a paragraph.

1 Identify the system, its form, and its function

Reminder

Form is what the system **is**. Function is what the system **does**, and a function is always a **process** acting on an **operand**. The operand is the thing whose state changes.

Worked example (Table 2.2 of the text)

System	Form	Process	Operand
Amplifier system	The amplifier circuit	amplifies	output signal
Design team	The team	develops	design

Your turn.

Do the same for one delivery.

System	Form	Process	Operand
The delivery system			

Then answer in one sentence each:

1. Why is the robot **not** the operand, even though the robot is the thing that moves?
2. What is the *instrument* of this function?

2 Identify the entities, the boundary, and the context

Below is a list of everything that might matter. Thinking this widely is called **holistic thinking**. Thinking this widely is also useless until you narrow it, which is called **focus**, and the rule of thumb is that you can only reason about 7 ± 2 things at once.

Candidate entities

delivery robot • cargo bin • phone app • dining hall worker • operations center operator • student • sidewalk • curb ramps • other pedestrians • campus wireless network • dining hall kitchen • grounds crew • the joystick • the battery • campus police

2a. Focus

Pick the five to seven entities that actually matter for getting one meal delivered. Cross out the rest, but keep the list, because you will need it in 2c.

2b. Form and function of each entity

For each entity you kept, write its form and its function, the same way the book does in Table 2.3. Function is still a process plus an operand.

Worked example (Table 2.3 of the text)

Entity form	Entity function
Sue	evaluates and approves the design
Resistor 1	sets the gain

Your turn.

Entity form	Entity function

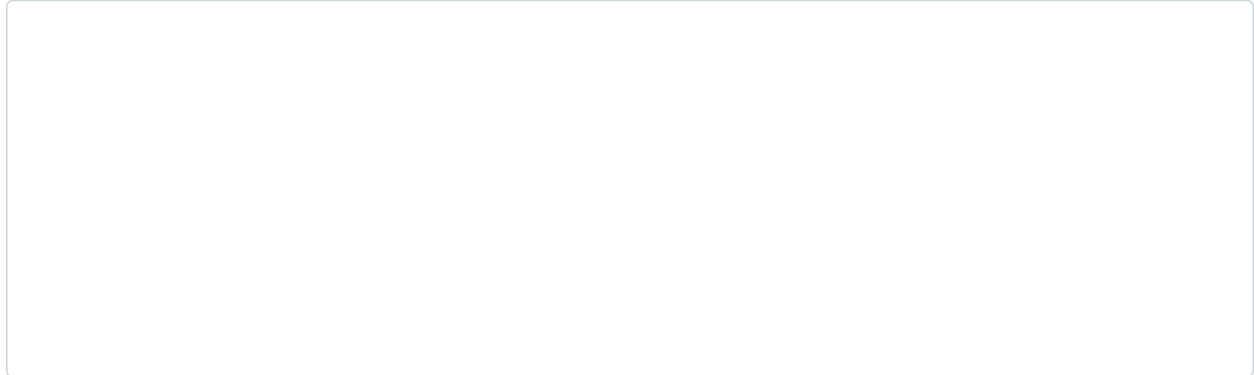
2c. Boundary and context

Draw a dashed line around the entities that are **inside** the system. Everything relevant that stays outside is the **context**. Then record which side of the line you put each of these on, and why.

Element	Inside or context?	Why (one sentence)
The student		
The sidewalk		
The curb ramps		

2d. The snow

In February, the snow problem happened at the edge of your diagram. Does your boundary make that failure easy to see, or does it hide it? There is no single right answer here, but say which one your boundary does.



3 Identify the relationships

Systems are entities *and their relationships*. The book splits relationships into two kinds.

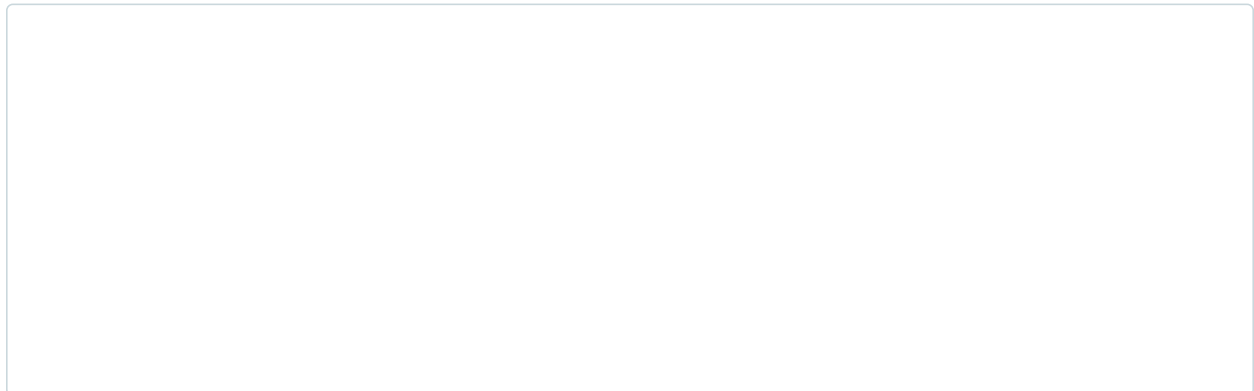
Reminder

Formal relationships are structure. They exist and sit still: a wire between two components, a person being on a team, two parts bolted together.

Functional relationships are interactions. Something is exchanged or transferred: a force, a signal, a message, an object.

Your turn.

Draw a relationship diagram like Figure 2.11 of the text: one box per entity, thin lines for formal relationships, arrows for functional interactions, and a dashed line for the system boundary.



Then answer:

1. Where a relationship crosses the boundary, the book calls it an **external interface**. List your external interfaces.
2. Pick one functional interaction and name what is being exchanged.
3. You kept about six entities. If you had kept all fifteen, roughly how many possible pairwise relationships would there be? (The book calls this the N^2 problem.) What does that tell you about why focus matters?

4 Identify the emergence

Emergence is what appears when the system operates and the parts are put together. The book sorts emergent properties into four boxes.

Worked example (Table 2.1 of the text)

	Anticipated	Unanticipated
Desirable	Cars transport people	Cars create a sense of personal freedom
Undesirable	Cars burn hydrocarbons	Cars can kill people

4a. Fill in all four boxes

Every one of the three surprises in the scenario belongs somewhere. Find a fourth entry yourself.

Anticipated	Unanticipated
Desirable	
Undesirable	

4b. Function or performance?

The book separates emergent **function** (what the system does) from emergent **performance** (how well it does it). Go back and label each of your four entries as function or performance.

4c. Where does the slowdown come from?

One sentence: the noon walkway slowdown is not caused by a broken part. Every robot and every student is behaving correctly. So where does the slowdown come from?

5 Thinking about complexity

5a. Which class is this system?

The three classes (Section 3.3 of the text)

Simple: one level of decomposition, no more than 7 ± 2 elements of form, and those elements are atomic parts.

Medium complexity: two levels, no more than about 81 parts.

Complex: looks like a two-level diagram, but at the second level you still have abstractions instead of parts.

Which is the delivery system? Defend it in two sentences. Look hard at your operations center entity before you answer.

5b. Atomic parts

Name two **atomic parts** in your diagram. The mechanical rule is: call it a part when you cannot take it apart.

5c. Complex is not the same word as complicated

Complex is a property of the system: many entities, many relationships. *Complicated* is about the observer: it overwhelms the person trying to understand it.

Was your Part 3 diagram complicated? What did you do, or what could you do, to keep it from being complicated without pretending the system is smaller than it is?